COUNTRY PROFILES

Uranium in Canada

UPDATED TUESDAY, 3 MARCH 2026

- Canada was the world's largest uranium producer for many years, accounting for about 22% of world output, but in 2009 was overtaken by Kazakhstan.
- Production comes mainly from the McArthur River (suspended since January 2018, restarted November 2022) and Cigar Lake mines in northern Saskatchewan province, which are the largest and highest-grade in the world.
- With known uranium resources of 694,000 tonnes of U_3O_8 (588,500 tU), as well as much continuing

Privacy - Terms

exploration, Canada has a significant role in meeting future world demand.

Canada is a country rich in uranium resources and a long history of exploration, mining and generation of nuclear power (for coverage of nuclear power, see information paper on Nuclear Power in Canada). To 2019, more uranium had been mined in Canada than any other country – 539,773 tU, about one-fifth of the world total. In 2020 it was overtaken by Kazakhstan/Uzbekistan (see also information page on Supply of Uranium).

Early uranium mining

In Canada, uranium ores first came to public attention in the early 1930s when the Eldorado Gold Mining Company began operations at Port Radium, Northwest Territories, to recover radium. A refinery to produce radium was built the following year at Port Hope, Ontario, some 5000 km away.

Exploration for uranium began in earnest in 1942, in response to a demand for military purposes. The strategic nature of such material resulted in a ban on prospecting and mining of all radioactive materials across Canada. In 1944, the federal government took over the Eldorado company and formed a new Crown corporation

which later became Eldorado Nuclear Ltd. Uranium exploration was restricted to the joint efforts of Eldorado and the Geological Survey of Canada.

Postwar, uranium exploration gathered pace when the wartime ban on private prospecting was lifted in 1947. Deposits around the Bancroft, Ontario, area were discovered by the early 1950s, and the first discovery in Ontario's Elliot Lake region was in 1953. The northern Saskatchewan uranium province was also discovered in the 1950s and Eldorado Nuclear began mining at Beaverlodge in 1953.

By 1956 thousands of radioactive occurrences had been discovered. Several proved to be viable deposits, and by 1959, 23 mines with 19 treatment plants were in operation in five districts. Of these 19, about 11 in the Elliot Lake area, including the largest plants, would come to be operated by Rio Algom and Denison Mines. Three other plants were located near Bancroft, three in northern Saskatchewan and two in Northwest Territories.

This first phase of Canadian uranium production peaked in 1959 when more than 12,000 tonnes of uranium were produced. The uranium yielded C$ 330 million in export revenue, more than for any other mineral export from Canada that year. However, this period marked the end of cost-plus production for export, and over the next few years the number of mines declined to four. Uranium production in the Bancroft area and at Beaverlodge ceased in 1982 and the last of the labour-intensive, lower-grade Elliot Lake mines closed in 1996.

During the 1960s the federal government supported the domestic uranium industry by initiating a stockpiling program which ended in 1974, after some 7000 tonnes of uranium were purchased at a cost of C$ 100 million. Uranium exploration was revived by expectations of nuclear power growth, and as a result several new uranium deposits were discovered in northern Saskatchewan's Athabasca Basin, starting in the late 1960s.

Second phase of uranium mining

A burst of exploration in the 1970s resulted in major discoveries in the Athabasca Basin in northern Saskatchewan, in Proterozoic unconformity deposits. Mines at Rabbit Lake, Cluff Lake and Key Lake started up in 1975, 1980 and 1983, which up until 2000 accounted for most of Canada's uranium production (14,223 tonnes of U_3O_8 in 1998). Cluff Lake, Key Lake and the original open pit at

Rabbit Lake have now been mined out (underground mining continued at Rabbit Lake to 2016). In 1988 the newly-formed Cameco Corporation* discovered the massive McArthur River deposit. Mines that began operation since 1999 now contribute most of Canada's production, notably McArthur River and Cigar Lake.

* In the late 1970s the Saskatchewan Mining Development Corporation, a provincial crown corporation, had taken a 20% interest in the Cluff Lake development and a 50% interest in Key Lake. In 1988 this merged with Eldorado Nuclear Ltd to form Cameco Corporation, now Canada's leading uranium producer.

Canada's Non-Resident Ownership Policy (NROP) for uranium projects had, since 1987, restricted foreign ownership of uranium mines to a maximum of 49%. The ownership structures then in place were "grandfathered" from the new policy and were able to continue. The policy provided for exemptions in situations where Canadian partners cannot be found, and applied to uranium production only. Uranium exploration was not subject to NROP, and there were several exploration-level uranium assets in Canada with major foreign ownership which cannot proceed to mining unless the NROP was liberalised. As part of the Canada-EU free trade agreement negotiated in October 2013, the foreign ownership restrictions would be relaxed. Foreign investment in Canada generally remains subject to the Investment Canada Act.

Current production

Canada's uranium production is tabulated below, and while relatively constant over the last few years, its share of world production dropped from about 20% to 15% before recovering to about 22% in 2016, worth about $2 billion. Over 85% is exported.

The majority of Canada's uranium resources are in high-grade deposits, some one hundred times the world average. Many of these have difficult mining conditions which require ground freezing.

The main uranium producers are Cameco and Orano Canada (formerly Areva Resources Canada). Cameco was formed in the 1988 merger of Saskatchewan Mining Development Corporation and the government-owned Eldorado Nuclear. The company issued its first public shares in 1991 and was fully privatized in 2002.

In the early 1990s, the Saskatchewan government had considered phasing out uranium mining in the province. This policy was later reversed after a joint Federal-Saskatchewan study panel on health, safety, environment and socio-economic impact found that the jobs provided by the industry would be hard to replace and that the environmental impact of mining could be minimized. Today, the provincial government actively supports uranium mining, and all new Saskatchewan uranium mines have international ISO 14001 environmental certification.

In December 2022 the Minister of Natural Resources released The Canadian Critical Minerals Strategy, which identified uranium as a critical mineral to cover industrial activities from geoscience and exploration to mineral processing, manufacturing and recycling applications, including support for research, development and technological deployment. A 30% critical mineral exploration tax credit introduced in April 2022 is intended to support specified exploration expenditures, applicable to specific critical minerals including uranium.

In January 2023 the Canadian Nuclear Safety Commission (CNSC) authorized a 20-year renewal of Cameco Fuel Manufacturing's operating licence for its facility at Port Hope in Ontario. The licence renewal also enables Cameco to increase the plant's annual production limit to 1650 tonnes of uranium (tU) – 24% higher than the limit in the previous licence, allowing the plant to operate at 100% capacity.

Annual uranium production (tonnes U_3O_8)[a]

	2013	2014	2015	2016	2017	2018	2019	2020	20
McArthur River	9135	8675	8673	8173	7303	90	0	0	0
Cigar Lake	0	156	5124	7863	8165	8166	8165	4581	553
McClean Lake	0	51	0	0	0	0	0	0	0
Rabbit Lake	1872	1889	1912	505	0	0	0	0	0
Total	11,007	10,771	15,709	16,541	15,467	8256	8165	4581	553
cf. World	69,966	66,087	71,113	74,357	71,361	63,861	64,554	56,287	56,

Domestic production in tonnes of uranium (as opposed to U_3O_8) is given in Note a.

McArthur River & Key Lake

The McArthur River uranium mine which commenced production in 1999 is one of the world's two largest (in terms of production) – the other being Cameco's Cigar Lake. It also has enormous reserves of high grade ore (16.5% U_3O_8 after allowance for dilution) located 600 metres underground. Remote control raise boring methods are used to mine the ore, which is then trucked 80 km south to be milled at Key Lake, site of the closed mine that once produced 15% of the world's uranium.

At the Key Lake mill, which has been modified for the McArthur River ore, the ore is blended with 'special waste rock' and processed to produce U_3O_8. Tailings are deposited in a mined-out pit. The licensed capacity of the Key Lake mill was 8,500 t/yr U_3O_8, but in 2015 CNSC approved an increase in mill production to 11,350 t/yr U_3O_8 and the same for the McArthur River mine itself. The production target was to reach 9908 tonnes/yr by 2018, but in November 2017 Cameco announced the temporary suspension of production from the McArthur River mining and Key Lake milling operations from January 2018. The suspension, due to persistent uranium price weakness, was expected to last for 10 months, but in July 2018, Cameco announced that the mine was shut for an "indeterminate period". Cameco earlier said that it would take 18-24 months for the mine to ramp back up to full production.

In February 2022 Cameco announced plans to return McArthur River to production, and 498 t U_3O_8 was produced in 2022. On 9 November 2022 it announced that uranium ore from McArthur River had been milled and packaged at Key Lake mill for the first time since these facilities were placed into care and maintenance in January 2018.

There has been no production from the Key Lake mine since 2002.

Cameco is the majority owner and operator of McArthur River mine (69.8%) as well as the Key Lake Mill (Orano is a 30.2% and 16.7% partner, respectively). Orano (then Areva Resources) earlier applied for a licence to process some McArthur river ore at McClean Lake.

Other deposits close to McArthur River are prospective.

McClean Lake

After starting operation in mid-1999, McClean Lake produced about 2500 t/yr of U_3O_8 from 2.4% ore up until 2005, although production was well down in 2006 through to 2010 due to lower ore grades. The mine was relicensed at 3640 t/yr. Operations have comprised three open pits, with an underground mine from Sue B pit planned for the future. McClean Lake also has high-quality new plant and infrastructure. It uses the first mined-out pit for tailings disposal – the JEB tailings management facility.

The McClean Lake mill has been upgraded and expanded to 8200 tonnes U_3O_8 (7000 tU) per year to accommodate all the ore from the Cigar Lake mine (see section on Cigar Lake below). Orano says that the mill is the most technologically-advanced in the world, being able to treat ore from less than 1% to 30% U, and is the only facility capable of processing high-grade uranium ore without diluting it. The CNSC gave regulatory approval at the end of 2012 to operate the mill with high-grade ore from McArthur River and increase production from 3640 to 5900 t/yr of concentrate. Orano (then Areva Resources) recommissioned the mill in 2013, and first deliveries from Cigar Lake arrived in March 2014. Areva Resources applied to increase the mill capacity to 10,900 tonnes U_3O_8 (9240 tU) per year, and in May 2016 CNSC licensed this. The licence was later renewed to 2027.

Efforts to increase production to fill the gap left by the delay in Cigar Lake production had limited success, and development of the nearby small Caribou deposit awaits improved economic conditions. Mining of Sue E deposit 2005-08 and Sue B in 2008 over 2008-10 provided ore for the mill until mid-2010 when it was shut down and put onto care and maintenance until it was upgraded to treat the Cigar Lake ore. Some 115,000 tonnes of low-grade ore remains stockpiled to be treated when markets improve. Reserves are small. In 2014 a little production was reported.

McClean Lake is majority-owned (70%) and operated by Orano Canada. Denison Mines (22.5%) and the Japanese company Overseas Uranium Resources Development (OURD Canada, 7.5%) are Orano's joint venture partners.

On 23 March 2020 Orano Canada decided to suspend production at the McClean Lake mill in response to the coronavirus pandemic. The decision was made in conjunction with Cameco's decision to suspend production at the Cigar Lake mine.

Mining resumed at the McClean Lake North deposit in June 2025 using the Surface Access Borehole Resource Extraction (SABRE) mining method to recover ore for processing at the McClean Lake

mill. SABRE, developed by Orano and Denison, is a non-entry, surface-based mining method that uses a high-pressure water jet placed at the bottom of a drill hole to excavate a mining cavity. The cuttings from the excavation process are then air lifted to the surface as a slurry for further processing. The SABRE mining method received a Canadian patent in February 2016 and a US patent in 2019.

Rabbit Lake

Uranium was discovered at Rabbit Lake in 1968 and it was brought into production by Cameco in 1975. Most of the deposit has been mined out, but reserves still exist at Eagle Point, where around 1700 t/yr of U_3O_8 from an ore grade of 2.1% have been mined underground in recent years. In 2016 production was suspended and the mine and mill transitioned to care and maintenance, which is expected to cost $35-40 million per year. Cameco was continuing its underground drilling reserve replacement program in areas of interest north and northeast of the current mine workings. However, in the second quarter of 2016 Cameco wrote off C$124 million as the full carrying value of the mine. It was North America's longest-producing uranium mine.

Remote control of a scoop tram at Rabbit Lake (Cameco)

Cigar Lake

Mining commenced at Cigar Lake in 2014. The proven and probable ore reserves at Cigar Lake are extremely large and very high grade. A 480-metre-deep underground mine was developed in very poor ground conditions – the orebody is actually in the soft Athabasca sandstone. Hence it uses ground freezing and remotely-controlled high pressure water jets at this level to excavate the ore. Production ramped up to 8200 t/yr U_3O_8 (7000 tU/yr) in 2017 and remained at that level in 2018 and 2019.

On 23 March 2020 Cameco decided to suspend production at Cigar Lake due to the threat of coronavirus. The decision was made in conjunction with Orano's McClean Lake uranium mill. Both facilities were be placed into care and maintenance. Cameco restarted production at Cigar Lake in September, and Orano restarted McClean Lake in the same month. However, both facilities were again suspended in December 2020 following a rise in cases in Saskatchewan.

Ore slurry from remote mining is trucked for toll treatment at Orano's expanded McClean Lake mill, 70 km northeast*. The McClean Lake mill was licensed only for 5900 t/yr, and in May 2016 Orano's (then Areva Resources) application to increase this to 10,900 t/yr was approved by the CNSC and in 2017 the licence was extended to 2027. Cameco reported that capacity in 2016 had been increased to 8200 t/yr U_3O_8. The mill is also expected to treat other ores in due course, notably from Midwest.

* Prior to October 2011 it was envisaged that all of the leaching would be done at McClean Lake and about half of the uranium solution would go onto Cameco's Rabbit Lake mill 70 km east for final production of uranium oxide concentrate. The revised arrangement reduced costs by 20%.

Cigar Lake headframes (Cameco)

Construction on the mine project began in 2005 with production originally scheduled to start in 2011. However, underground floods in 2006 and 2008 set the start date back until 2014 and increased the overall cost of the project from C$660 million to about C$2.6 billion. In February 2010, dewatering was completed and remediation proceeded. The 425 m level was backfilled and new workings developed in more competent rock at 480 m level. The first jet boring commenced in December 2013. The estimated average cash operating cost for Cigar Lake increased from $14.40 per pound U_3O_8 in 2007 to $23.14, but revised milling plans reduced this estimate to $18.60 per pound. The first ore slurry was sent to the McClean Lake mill in March 2014, and treatment began there in October 2014. Initial treatment of the ore is at Cigar Lake, with average slurry grade of about 15% U_3O_8 for shipment to McClean Lake.

Some 1.3 million cubic metres of waste rock from Cigar Lake is being emplaced under water in the Sue C pit at McClean Lake, to prevent acid generation from it. Tailings will remain at McClean Lake.

A Cigar Lake II deposit nearby is being investigated, as is the Waterbury Cigar Lake project as a possible northern extension of the Cigar Lake deposit.

Cameco, which has 54.547% ownership, is managing the joint venture, with Orano holding 40.453% and TEPCO Resources 5%. Japanese company Idemitsu Canada Resources had held

7.875% but divested in May 2022, with Cameco and Orano acquiring their shares.

Resources

Canadian uranium reserves and resources[c]

Mine	Province	Operator	tonnes U	tonnes U_3O_8	Average ore grade U_3O_8[d]	Category
McArthur River	Sask	Cameco	142,000	167,700	9.60%	proven & probable reserves
			1850	2180	3.8%	measured & indicated resources
Cigar Lake	Sask	Cameco	82,720	97,550	15.9%	proven & probable reserves
			32,500	38,340	16.24%	measured & indicated resources
Millennium	Sask	Cameco	29,200	34,400	2.39%	indicated resources
			11,150	13,160	3.19%	inferred resources
Rabbit Lake	Sask	Cameco	15,270	18,000	0.79%	indicated resources
McClean Lake	Sask	Orano	284	335	0.38%	proven & probable reserves
			5903	6961	0.57%	measured & indicated resources
Midwest	Sask	Orano	19,500	23,000	2.3%	indicated resources
Dawn Lake	Sask	Cameco	6885	8120	4.42%	indicated resources
Wheeler River Phoenix & Gryphon	Sask	Denison	23,000	27,000	16.2%	probable resources
			19,000	22,000	1.5%	probable resources

Mine	Province	Operator	tonnes U	tonnes U_3O_8	Average ore grade U_3O_8[d]	Category
Fox Lake	Sask	Cameco	26,195	30,892	7.99%	inferred resources
Shea Creek	Sask	Orano-UEX	26,100	30,770	1.48%	indicated resources
			10,870	12,800	1.01%	inferred resources
Roughrider*	Sask	Hathor/Rio	22,300	26,300	2.0-11.6%	indicated & inferred resources
Tamarack	Sask	Cameco	6900	8100	4.42%	indicated resources
Patterson Lake South	Sask	Fission	39,900	47,100	1.85%	indicated resources
			12,600	14,900	1.30%	inferred resources
Arrow	Sask	NexGen	80,600	95,000	4.35%	measured resources
			18,150	21,400	1.36%	indicated resources
			31,000	36,700	0.83%	inferred resources
Kiggavik	Nunavut	Orano	48,953	57,730	0.554%	indicated resources
Michelin	Labrador	Aurora (Paladin)	32,430	38,240	0.10%	measured & indicated resources
			8820	10,400	0.12%	inferred resources
Jacques Lake	Labrador	Aurora (Paladin)	4000	4700	0.08%	measured & indicated resources
Matoush	Quebec	Strateco/ICU	4740*	5590	0.954%	indicated resources
			6320	7450	0.442%	inferred resources

* Not included in latest company reports.

Proposed mines

Midwest

In March 2018 Denison announced that indicated resources were 23,000 tonnes of U_3O_8 at 2.3%, 18,100 tonnes of this in the Main deposit at 4.0% grade. Inferred resources were 8250 t U_3O_8 at 1.1%. The Midwest A deposit, 2km to the north, had 4900 t U_3O_8 indicated at 0.87%.

The original plans were for an underground mine, utilising ground freezing and water jet boring, but current plans call for a large open pit mine that will go to a depth of 215 metres and involve draining an arm of South McMahon Lake. The ore will be shipped 15 km to the McClean Lake mill. A comprehensive environmental assessment for the project began in 2006 and federal environmental approval for open pit mining was received in August 2012. Other potential mining methods are being evaluated, including conventional underground and surface jet bore drilling, using the SABRE ("Surface Access Borehole Resource Extraction") mining technology.

Production was originally scheduled to begin in 2011, but in 2008 the starting date was postponed due several factors, including a 50% rise in the initial estimated capital costs of $435 million. The Midwest project is being managed by Orano, which owns 69.16%. Denison Mines has a 25.17% stake and OURD Canada 5.67%. Denison says it needs a $60/lb uranium price to justify development.

Dawn Lake

Although its development is much further off, a deposit of more than 8000 tonnes U_3O_8 of indicated resources is prospective at Dawn Lake in northern Saskatchewan. Grades of up to 30% ore at depths of 280 metres have also been reported nearby. Cameco has 57.4%, Orano 23.1% and Japan-Canada Uranium subsidiary JCU (Canada) Exploration 19.4%.

Cameco's Tamarack deposit associated with Dawn Lake has an indicated resource of 8100 tonnes U_3O_8 at 4.42%, requiring underground mining.

Millennium

The Millennium deposit (now 70% owned by Cameco, 30% JCU) has indicated resources of 23,100 tonnes of 4.5% grade U_3O_8 and 7575 tonnes of 2.1% grade inferred resources, in basement rock, below the soft sandstone. It is between McArthur River and Key Lake, and ore would be milled at Key Lake. A feasibility study on the project led to Cameco seeking approval to mine it at about 2500 tU/yr. The environmental assessment was approved at the end of 2013. Underground development was envisaged over 2013-17, but in mid-2013 Cameco said it was not a primary project, and in May 2014 it halted developments pending improvement in the uranium prices. In 2012 Cameco paid C$150 million for Orano's (then Areva Resources) 28% share.

An exploration camp at Millenium (Cameco)

Kiggavik

In the Kivalliq region of Nunavut Territory, some 500 km north of Manitoba, a joint venture headed by Orano has been conducting a feasibility study on the Kiggavik uranium deposit in the Thelon Basin, with 48,950 tU indicated resources at 0.47%U grade. It is 80 km west of the community of Baker Lake. The indigenous Inuit

organization, Nunavut Tunngavic, reversed its previous ban on uranium exploration and mining in 2006, but the project has faced opposition from other groups. In March 2010, the Nunavut government ruled that the proposal would be reviewed by a territorial regulator rather than undergo a federal environmental assessment.

In October 2014 Orano (then Areva Resources) submitted a final environmental impact statement to the Nunavut Impact Review Board. The project involves the development of three open pit mines at Kiggavik and both an open pit mine and an underground mine at Sissons, with tailings placed in a mined-out pit. A dock site would be at Baker Lake.

Orano and its partners, JCU (Canada) Exploration (33.5% in Kiggavik) and Daewoo (1.7%), hope for a start-up of the mine and mill complex when the market improves, to produce about 3000 tU/yr over 14 years.[e] In May 2015 the Nunavut Impact Review Board declined to approve the project due to the indefinite start date, but invited resubmission when Orano (then Areva Resources) could provide a more definite timescale. Orano (then Areva Resources) has protested to the federal minister responsible, saying that the process has been lengthy and thorough and the lack of firm start date should not prevent approval. In July 2016 the Canadian government agreed with the Nunavut Impact Review Board recommendation, and said Orano (then Areva Resources) could resubmit when it had a proposed start date.

Michelin

The Michelin deposit is in Eastern Canada's Central Mineral Belt, in Labrador. It is being drilled in a C$21million programme by Aurora Energy Resources (subsidiary of Paladin Energy, acquired in 2011). In August 2018 Paladin announced that EDF Claimants held a 50% participating interest in the project, following resolution of legal proceedings.

Michelin and nearby **Jacques Lake** are the main deposits, with minor amounts in Rainbow and three others. All are metasomatite-type mineralization except for Moran Lake which is iron-ore-copper-gold (IOCG) with subeconomic uranium. Michelin has rare earths. In 2009, a positive economic assessment of the project proposed investment of US$ 984 million to set up mine and mill, with production ramping up to 3000 t/yr. However, in 2015 the Labrador projects were suspended due to low uranium prices.

The Michelin deposit has measured resources of 15,490 tonnes U_3O_8 (13,135 tU), indicated resources of 22,750 tonnes U_3O_8 (19,290 tU) and inferred resources of 10,400 tonnes U_3O_8 (8820 tU) based on NI 43-101 figures published in mid-2014. About 40% of the measured and indicated resource in Michelin is amenable to open cut mining. Measured and indicated resources in five other associated deposits, mostly Jacques Lake, are 7500 tonnes U_3O_8.

A Nunatsiavut government three-year moratorium had been in place until March 2011, affecting Michelin, and expiry of this coincided with completion of a land use planning assessment undertaken jointly by the Nunatsiavut and Newfoundland-Labrador governments. After establishing a lands administration system, developing environmental protection legislation, and following a review and public consultation, in December 2011 the Nunatsiavut Assembly voted unanimously to lift a moratorium on the development of uranium deposits on Labrador Inuit lands, and this was legislated in March 2012. Five of Aurora's six uranium deposits in the Central Mineral Belt fall within the Labrador Inuit lands. In June 2015 the Canadian government approved Paladin's ownership of the project, exempting it from the Non-Resident Ownership Policy (NROP) applying generally, allowing it to proceed to production.

Wheeler River: Phoenix & Gryphon

Denison's Wheeler River project is halfway between Key Lake and McArthur River and immediately east of Millennium. It is along strike from McArthur River and geologically very similar, with some high-grade uranium mineralisation. A September 2018 pre-feasibility study said that the project was expected to produce 42,000 tU over a 14-year mine life, with Phoenix being mined by in situ leaching and Gryphon ore being milled at McClean Lake, 120 km away.

In January 2018 the NI 43-101 compliant indicated resources for the **Phoenix** deposits were confirmed as 27,000 tU at an average grade of 16.22%U, with cut-off grade 0.68%U. The September 2018 pre-feasibility study gave probable reserves of 23,000 tU for Phoenix at 16.2%U. All-in cost for Phoenix production was $11.57/lb (about US$ 9.00/lb). In December 2018 Denison announced that it would proceed with developing an ISL operation at Phoenix with a view to production in 2024. It will have first-of-a-kind freeze wall confinement for leach solution. The CNSC approved a licence for Phoenix in February 2026 and the same month Denison made the final investment decision to develop it with

construction slated to start in March 2026 and first production by mid-2028. Denison said that a planned 11-year lifespan for Phoenix could produce 25,800 tU.

The **Gryphon** deposit discovered in 2014 about 3 km northwest of Phoenix was upgraded to 23,860 tU indicated resources at 1.44%U, with cut-off grade of 0.17%U. Inferred resources at both deposits added 1360 tU. In September 2018 a pre-feasibility study gave probable reserves of 19,000 tU for Gryphon at 1.5%U. All-in cost for Gryphon was $29.67/lb (about US$ 23.10/lb).

After taking over Cameco's 26.7% interest in 2018, Denison has a 90% interest and JCU (Canada) 10%.

Exploration prospects

In addition to mining operations planned for the near future, active exploration involving more than 40 companies continues in many parts of Canada. While exploration has concentrated on northern Saskatchewan, new prospects extend to Labrador and Nova Scotia in the Atlantic provinces, Quebec province, Nunavut Territory in the far north, and Ontario's Elliott Lake area. Resource figures quoted are generally NI 43-101 compliant.

The 2014 IAEA *Red Book* says that in 2011-12 "uranium exploration remained focused on areas favourable for the occurrence of deposits associated with Proterozoic unconformities in the Athabasca Basin of Saskatchewan, and to a lesser extent, similar geologic settings in the Thelon and Hornby Bay basins of Nunavut and the Northwest Territories."

However, three provinces, none of which are familiar with uranium mining, have banned uranium mining and exploration: Nova Scotia (since 1981), British Columbia (1980-87 and from 2008) and Quebec (since 2013). Tailings management concerns appear to be the reason.

In Nunavut, Kivalliq Energy (part of Aurora Group) has identified 19,680 t U_3O_8 (16,690 tU) inferred resources grading 0.69% U_3O_8 with 0.2% cut-off in its Lac 50 Trend deposit at its Angilak project. This includes 12,730 tonnes in Lac Cinquante deposit and 6,950 tonnes in J4/Ray, with good intersections in Dipole, 25 km southwest of it, still unquantified. Also in Nunavut, at Amer Lake, Uranium North Resources has reported inferred resources of 9500 t U_3O_8.

In uranium-rich northern Saskatchewan, exploration projects are now well-advanced at several locations.

In 2016 Cameco reported on its Fox Lake deposit on its **Read Lake** property near McArthur River, with 30,900 t U_3O_8 @8% as inferred resources. Orano holds 22%. Also Cameco has signed a farm-in agreement with CanAlaska Uranium for its West McArthur project, adjacent to Fox Lake and 15 km west of the McArthur River mine. This became a joint venture in December 2018, with CanAlaska 70% owner and also operator.

The **Shea Creek** project (one of nine JVs 51% owned by Orano, 49% UEX Corp. which is 21.3% owned by Cameco) in the western Athabasca Basin 13 km south of Cluff Lake has reported high-grade ore. In April 2013, UEX announced indicated resources of 30,770 t U_3O_8 grading 1.48% and inferred resources of 12,800 tonnes grading 1.01%, as of January, with cut-off 0.30%. The deposit remains open. Production at about 2500 tU/yr is envisaged. Exploration expenditure to the end of 2012 was C$ 40.5 million, and actively continues in 2015 though with no change in resource figures.

UEX is also exploring the Horseshoe and Raven deposits at **Hidden Bay** in the eastern Athabasca basin (5 km from Rabbit Lake and 12 km from McClean Lake). The Horseshoe deposit has indicated resources of 10,700 tonnes of U_3O_8 at a grade of 0.215% at 100 to 450 m depth. Raven has indicated resources of 6300 tonnes at 0.117%, with cut-off 0.05%, at 100 to 300 m deep (2021 figures). These amounts increase slightly with 0.02% cut-off. A preliminary technical assessment of the deposits in 2011 was positive and recommends a preliminary feasibility study which also includes the smaller but shallow West Bear deposit (720 t at 0.91%). The 2011 report assumes Horseshoe access by decline and Raven by open cut, with toll milling and tailings management at Rabbit Lake mill over seven years. UEX is also considering heap leaching the ore.

Another UEX project is Christie Lake, 9 km northeast of MacArthur River and along strike, where some high-grade intersections have been made. UEX has 30% with JCU (Canada) Exploration and is moving to 70%. In December 2018 UEX announced a maiden inferred resource of 7830 tU at 1.57% U_3O_8 (1.33%U).

With the Korea Waterbury Uranium Limited Partnership led by Korea Electric Power Corp (Kepco), Denison (65.45%) is exploring the **Waterbury Lake** area near Midwest. In September 2013 it announced NI 43-101 indicated resources of 4900 tU grading 1.7%U for the **J-Zone** at Waterbury Lake, since renamed Tthe

Heldeth Túé (THT)*. This is along strike and about 2km from Midwest A deposit and contiguous with Roughrider. In 2020 Denison announced an assessment of Waterbury Lake for ISL mining, with processing at McClean Lake mill, to produce 3730 tU over six years. Denison also is investigating its Jasper Lake project, 40 km east of Cigar Lake.

* Tthe Heldeth Túé is a Dené name pronounced “Tey Hel-deth Tway”

Orano reports that its and Cameco's Waterbury Cigar Lake project aims to discover an extension of the nearby Cigar Lake orebody. Equity is the same as for Cigar Lake mine.

ALX Uranium Corp is farming into Orano Canada's Close Lake Uranium Project in eastern Athabasca Basin, with the right to reach 51%. Orano currently has 74.4%, Cameco 15% and JCU (Canada) 10.6%.

Fission Uranium Corp is exploring **Patterson Lake South** on the southwest margin of the Athabasca Basin, 90 km south of Cluff Lake, incorporating the Triple R deposit. Indicated resources for the project in October 2018 were 39,300 tU, 80% of which were mineable by open pit and graded 1.97%U. Inferred resources were 12,600 tU, mostly accessible only underground, graded 1.04%U. In April 2019 probable reserves were 31,650 tU at 1.37%U accessible by open pit (NI 43-101 complaint). The reserves include a substantial high-grade zone. Most of the deposit is less than 250m deep.

A prefeasibility study reported in April 2019 envisaged a hybrid open-pit and undergound operation producing an average 5700 tU per year over the first five years. Average operating costs are estimated at US$6.77 per pound U_3O_8 over the life of mine. An alternative scenario with all mining underground is economically attractive but reduces production from 35,000 tU in the base case to 31,300 tU. A new Patterson Lake South mill could potentially serve the Western Athabasca Basin.

The company was spun out of Fission Energy Corp after Denison bought it in 2013, and it then took full ownership of the Patterson Lake prospects, paying Alpha Minerals C$ 185 million for its half share. In July 2015 it announced a merger with Denison Mines, to become Denison Energy Corp, but this was subsequently aborted after failure to secure agreement from Fission shareholders. In December 2015 China’s CGN Mining Company Ltd agreed to buy 20% of Fission for C$82 million. An offtake agreement will entitle

CGN to up to 35% of Patterson Lake South production at a 5% discount on prevailing spot market prices.

Fission 3.0 has the Clearwater West project adjacent to Patterson Lake South, and Canex Energy is farming into this.

NexGen Energy is drilling its **Rook 1** property in the southwest Athabasca basin near Patterson Lake South and along strike from it, with basement-hosted mineralization. The **Arrow** deposit has 95,000 tonnes U_3O_8 (86,000 tU) measured resources at average 3.7%U, mostly in a very high-grade part of it grading 16.1%. Indicated resources are 18,000 tU at 1.15%U (NI-43-101, 2019). It is basement-hosted in igneous rock starting 100 m deep and extending down for 800 m. A feasibility study early in 2021 put the capital cost of the mine, plant and infrastructure at C$1.3 billion, for average annual production of 11,000 tU in years 1-5, but with very low operating cost. All tailings will be disposed of underground. Development of it is envisaged by about 2025. The Bow prospect is 3.7 km northeast of Arrow and very close to Purepoint's Spitfire.

Purepoint Uranium Group is drilling its **Hook Lake** prospect, along the Patterson Lake trend in southwest Athabasca Basin, in JV with Cameco and Orano (39.5% each), and has some positive intersections, including Spitfire.

Denison is drilling its Hook-Carter prospect in the Patterson Lake corridor. ALX Uranium holds a 20% share.

The **Roughrider** prospect 24 km from Rabbit Lake in Athabasca Basin at the time of takeover of Hathor Exploration by Rio Tinto had inferred resources of 13,700 t U_3O_8 at 11.58%, with 0.4% cut-off in the Eastern zone, for underground mining, and in the West zone indicated resources of 7800 t U_3O_8 at 1.98% and 4800 t inferred resources at 11.03% with 0.5% cut-off, for open pit mining. The East Zone is a series of moderately-dipping stacked, parallel lenses (greater than 0.5% U_3O_8). Rio Tinto does not list any figure in its 2015 report. It said: "Orebody knowledge progressed with geological and structural interpretation and resource modelling" and recorded an impairment charge of US$ 229 million, valuing the project at US$ 250 million. A preliminary economic assessment for Hathor had suggested low production costs over an 11-year mine life producing 1900 tU per year. Hathor was subject to a takeover bid from Cameco but agreed to another from Rio Tinto, valuing the company then at C$ 654 million.

In Nova Scotia, exploration has been proposed at Millet Brook, but it awaits a review of a 1985 moratorium on uranium mining in the province.

In Quebec, uranium exploration is underway at several locations with a total of more than 40,000 tonnes of indicated or inferred deposits. However, in April 2013 the Quebec government announced that no permits for uranium exploration or mining would be issued in Quebec until an independent study into its environmental impact had been completed. In addition to environmental groups, the Grand Council of the Crees is opposed to any uranium mining in Quebec. A government decision was expected in mid-2015.

A 626-page report by Québec's *Bureau d'audiences publiques sur l'environnement* (BAPE) was published by the province's minister for sustainable development, environment and climate change in July 2015. It followed one year's work by a commission set up by BAPE in May 2014 to study the environmental and social impacts of uranium exploration and mining and conduct public hearings. The report expresses concern about managing mining wastes. While concluding that it would be "premature" to authorize the development of a uranium industry now, the BAPE commission urged the Québec government not to preclude uranium mining on a temporary or permanent basis because of potential legal and economic impacts. The government would need to ensure social acceptability through an extensive information programme and cooperation and consensus-building strategy; overcome "technological uncertainties and current gaps in scientific knowledge"; and develop a legal framework to allow it to control uranium mining operations in the province. The head of CNSC then wrote to Quebec's minister questioning the report's recommendations, saying that they lacked "scientific basis and rigour" and hence were misleading for all Canadians. "To suggest that uranium mining is unsafe is to imply that the CNSC and the government of Saskatchewan have been irresponsible in their approval and oversight of the uranium mines of Canada for the last 30 years." "It is clear that the BAPE's recommendations not to proceed is based on the perceived lack of social acceptance and not on proven science."

In the Otish Mountains of central Quebec Strateco Resources had been granted a licence by the CNSC to conduct underground exploration on the **Matoush** deposit from 2014, and commenced environmental studies for the project. Matoush has indicated resources of 5600 tU at 0.81%U and inferred resources of 6320 tU at 0.375%U, and the company projected mine production of 1000 tU/yr over seven years from 2016.

Following the April 2013 moratorium, in November 2013 the Quebec government refused to authorize the Matoush underground

exploration phase. Strateco then commenced legal action against the provincial government.* It said it had invested over $123 million in the project to date, and announced an impairment charge of $87 million in its accounts due to its inability to proceed with the project's underground exploration program, the suspension of exploration and evaluation planned for 2014, and the uncertainty created for Quebec's uranium industry.

* On 5 December 2013, Strateco filed a motion to invalidate the decision rendered by Quebec's Minister of Sustainable Development, the Environment, Wildlife and Parks, whereby he refused to deliver the certificate of authorization for the underground exploration phase of the Matoush project.

In December 2014 Strateco launched a C$ 183 million claim against the provincial government in the Superior Court of Quebec for the loss of its investments. "It should be recalled that on the basis of extremely detailed, rigorous environmental and social impact studies, Strateco received approvals for the underground exploration phase of the Matoush project from the Canadian Nuclear Safety Commission, the federal Minister of the Environment and the federal administrator of the James Bay and Northern Quebec Agreement, as well as a positive recommendation from the provincial evaluation committee," the company said. In February 2016 it added $10 million punitive damages to the claim, which was heard in February 2017. In June 2017 the claim was rejected by the court.

In November 2014 Toro Energy from Australia acquired a 19.8% interest in Strateco as part of a financing package. In June 2015 Strateco filed for bankruptcy protection in the Superior Court of Quebec under the Companies Creditors Arrangements Act. In May 2021 International Consolidated Uranium agreed to buy the project from Strateco.

Abitex Resources/ABE Resources is exploring its **Epsilon** project in the Otish Mountains of Quebec. Azimut Exploration has committed C$42 million to uranium exploration, mainly for the **Katavic** project in Quebec's northern Nunavik region and other prospects in the Ungava Bay region further north. Uracan Resources reports 3100 tonnes U_3O_8 of indicated resources and 16,900 tonnes of inferred resources in the Double S zone at its North Shore prospect in eastern Quebec. Orano is establishing a joint venture with Waseco Resources to explore the Labrador Trough project.

In Northwest Territories, Cameco has the prospective **Boomerang** project in the southwest Thelon Basin. Land access issues hinder

active exploration at present.

The Elliot Lake area of Ontario, which was the centre of Canada's early uranium mining, is again attracting exploration. In September 2008, Pele Mountain Resources commenced the permitting process for its **Eco Ridge** underground uranium and rare earth oxides mine and processing facility in the region. Eco Ridge contains indicated resources of 10,250 tonnes U_3O_8 and inferred resources of 17,100 tonnes U_3O_8 along with significant REO resources. The Serpent River-Pecors deposit is a few kilometres east.

In British Columbia, the **Blizzard** prospect south of Kelowna, which was first explored in the 1980s, was revived by Boss Power. The company challenged a provincial government moratorium on exploration and mining imposed in April 2008, and the British Columbia government settled by paying the company $30.36 million in 2014.

Legacy mine rehabilitation

The Saskatchewan Research Council (SRC) is undertaking remediation of several mine and mill sites in northern Saskatchewan including Gunnar mine (operated 1955-64) and Lorado mill as well as 35 satellite mine sites near Lake Athabasca. This is under Project CLEANS (Cleanup of Abandoned Northern Sites) – a multi-year project funded by the governments of Saskatchewan and Canada. The estimated cost for Gunnar is C$ 280 million. In the 1960s and 1970s most northern sites were subject to minimal clean-up and rehabilitation and the operating companies are long defunct.

Similar work has been undertaken around Elliott Lake in Ontario, attending to 12 mines and ten mill sites, though many of these were rehabilitated in the 1980s-90s by companies which continued in business, notably Rio Algom and Denison.

Recent transfers to foreign ownership

As well as foreign equity in the companies with uranium mines, in recent years there has been increased interest in exploration companies. Some companies active in Canada are foreign-based, e.g. Orano. The following table outlines some recent foreign investment in Canadian-based or established explorers, or particular projects, which have credible resources.

Company	Main deposit	Overseas investor and share	Value of share	When
Uranium One	In Kazakhstan, also USA	ARMZ, 51.4%	C$1425 million	Dec 2010
Hathor Exploration	Roughrider	Rio Tinto, 100%	C$654 million	Nov 2011
Aurora Energy	Michelin	Paladin Energy 100%		2011

Non-proliferation

Canada's uranium is sold strictly for electrical power generation only, and international safeguards are in place to ensure this. See fuller statement in information paper on Nuclear Power in Canada.

Canada is a party to the Nuclear Non-Proliferation Treaty (NPT) as a non-nuclear weapons state. Its safeguards agreement under the NPT came into force in 1972 and the Additional Protocol in relation to this came into force in 2000.

Notes & references

Notes

a. Data: company sources

Annual uranium production (tonnes U)

	2013	2014	2015	2016	2017	2018	2019	2020	202
McArthur River	7744	7356	7354	6928	6193	76	0	0	0
Cigar Lake	0	132	4345	6666	6925	6925	6925	3885	469
McClean Lake	0	43	0	0	0	0	0	0	0
Rabbit Lake	1587	1602	1621	428	0	0	0	0	0
Total	9331	9134	13,320	14,022	13,116	7001	6925	3885	469
cf. World	59,331	56,041	60,304	63,207	60,514	54,154	54,742	47,731	48,

[Back]

b. Data: company sources. Where an asterisk (*) is shown, the figures are from the World Nuclear Association Market Report. [Back]

c. Data: company sources. In Canadian figures resources do not include reserves and are reported in accordance with Canadian standard NI-43-101. [Back]

d. Average ore grades given as percentage of U_3O_8 in the ore. [Back]

e. The two parts of the project (Kiggavik and Sissons) are operated by Areva Resources Canada Inc.; Sissons was held 50% by Areva in joint venture with JCU (Canada) Exploration Co. Ltd. (48%) and Daewoo Corporation (2%); and Kiggavik itself was held 99% by Areva and 1% by Daewoo. However, in 2014 Mitsubishi showed 33.5% of Kiggavik being held by JCU, an equal three-way JV among Mitsubishi Corp, Itochu and OURD. [Back]

General sources

Uranium webpage on Natural Resources Canada website (www.nrcan.gc.ca)
Cameco annual reports
Uranium in Saskatchewan series of fact sheets available on Cameco website (www.cameco.com)
Areva Resources website (www.cri.ca)
Canadian Nuclear Association website (www.cna.ca)
Uranium mines and mills waste section of Canadian Nuclear Safety Commission website (www.nuclearsafety.gc.ca)
OECD Nuclear Energy Agency and International Atomic Energy Agency, 'Red Books'

Appendix

Appendix 1: Brief History of Uranium Mining in Canada

Appendices

Uranium in Canada Appendix 1: Brief History of Uranium Mining in Canada

Related information

Canada: Nuclear Power